Total No. of Questions : 5] [Total No. of Printed Pages : 5

Roll No. ____________________

REF NO. 052708

B. Sc. (Hons) Semester V Examination
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PHYSICS

Paper No. BPT-505 : Electromagnetic Theory

Time : 3 hours **Full Marks : 70**

(Write your Roll No. at the top immediately on the receipt of this question paper)

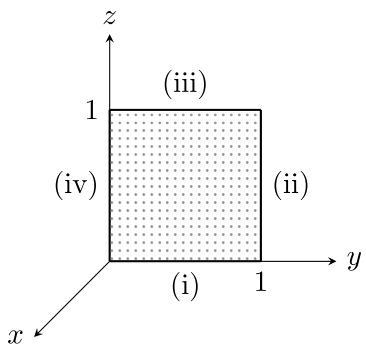
The figures in the right-hand margin indicate marks.
Answer all questions.
All symbols have their usual meanings unless specified otherwise.

1. **Answer any four of the following :** **4×5**

(a) For the vector

$$\vec{v} = (2xz + 3y^2)\hat{a}_x + (4yz^2)\hat{a}_z$$

calculate the line integral for the curve (iv) in the given figure:



(b) If $\hat{r}$ is the unit vector defined as $\hat{r} = \vec{r}/r$, $r = \sqrt{x^2 + y^2 + z^2}$. Compute $(\vec{r} \cdot \vec{\nabla})\hat{r}$.

(c) Show that Maxwell's equations in electrodynamics are consistent with continuity equation.

(d) Find the skin depth of copper with $\sigma = 5.8 \times 10^7$ mho/m, $\mu = \mu_0$ at frequency 10^8 Hz.

(e) In a non-magnetic medium $\vec{E} = 4\sin(2\pi \times 10^7 t - 0.8x)\hat{a}_z$ V/m. Compute the time average power carried by the wave.

(f) In a lossy dielectric medium, show that $\frac{\theta}{2} = \theta_\eta$, where θ is the loss angle of the medium and θ_η is the phase of the intrinsic impedance η.

2. (a) Consider the multipole expansion of the potential in electrostatics and show that the 'dipole term' can be written in the form $V = \frac{1}{4\pi\varepsilon_0}\frac{\vec{p}\cdot\hat{r}}{r^2}$, where $\vec{p}$ is the dipole moment. **6**

   (b) Consider the Ampere's law in the presence of magnetic materials. Write it in both differential and integral forms. Obtain the magneto-statics boundary condition using the integral form of the equation. **6**

3. (a) Discuss the propagation of e.m. wave in lossy dielectric medium and derive the expression for the magnitude of the intrinsic impedance of the medium. **7**

   (b) In a lossless medium for which $\eta = 60\pi$, $\mu_r = 1$ and

$$\vec{H} = -0.1\cos(\omega t - z)\hat{a}_x + 0.5\sin(\omega t - z)\hat{a}_y \text{ A/m}.$$

   Calculate the angular frequency (ω). **5**

4. Obtain the electrostatic potential both inside and outside of a hollow sphere which is maintained at a constant surface charge density σ_0. **12**

Or

   (a) What do you mean by divergence of a vector? Calculate the divergence of

$$\vec{v}_0 = y^2\hat{a}_x + (2xy + z^2)\hat{a}_y + 2yz\hat{a}_z.$$

5

   (b) Explain the meaning of the gauge transformation. Write the Lorentz gauge condition and show that both electrostatic and vector potentials obey the wave equation in the Lorentz gauge in the absence of sources. **7**

5. (a) Show that TM_{11} is the lowest order mode of all the TM_{mn} modes supported by the rectangular waveguide. **7**

(b) Derive the relation between reflection and transmission coefficients for an obliquely incident e.m. wave in which wave electric field is parallel to the plane of incidence. **7**

Or

(a) A plane e.m. wave normally incident on the boundary $z = 0$ formed by a lossless dielectric medium and a good conductor. Obtain the z-values at which minimum value of $|E_1|$ occurs in the medium 1, where E_1 is the total electric field in medium 1. **8**

(b) Show that a TE_{10} mode of a rectangular waveguide can be represented as a sum of two plane TEM waves propagating along zigzag paths between the guide walls. **6**

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